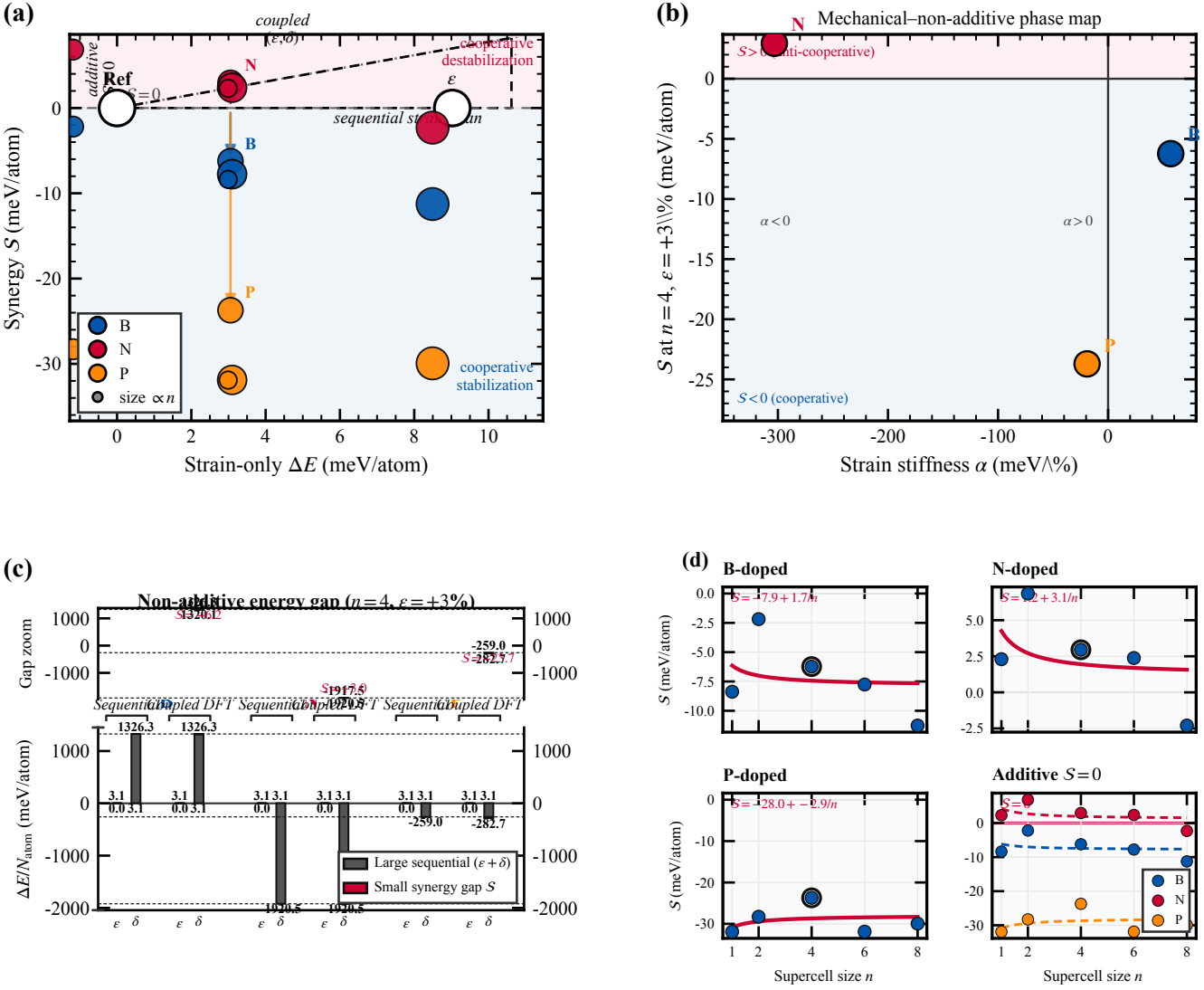

(a)
coupled
(ε,δ)
additive
S = 0
Ref
N
ε
B
P
cooperative
destabilization
sequential stabilization
cooperative
stabilization
B
N
P
size ∝ n
Synergy S (meV/atom)
Strain-only ΔE (meV/atom)
(b)
Mechanical–non-additive phase map
N
S > 0 (anti-cooperative)
B
α < 0
α > 0
P
S < 0 (cooperative)
S at n = 4, ε = +3\\% (meV/atom)
Strain stiffness α (meV/Å%)
(c)
Non-additive energy gap (n = 4, ε = +3%)
Gap zoom
ΔE/N_atom (meV/atom)
Sequential
Coupled DFT
1326.3
-259.0
-282.7
-1920.5
Large sequential (ε + δ)
Small synergy gap S
ε
δ
(d)
B-doped
S = −7.9 + 1.7/n
N-doped
S = 2 + 3.1/n
P-doped
S = −28.0 + −2.9/n
Additive S = 0
S (meV/atom)
Supercell size n
B
N
P